BIDIRECTIONAL LIQUID NEURAL NETWORKS (BLIQNET) *

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ABSTRACT

Standard feedforward and continuous-time deep learning models suffer from catastrophic mode collapse and topological tearing when modeling ill-posed inverse problems and self-intersecting manifolds. In many-to-one mappings, deterministic vector fields are forced to average valid targets, failing to capture the true underlying data distribution. We introduce Bidirectional Liquid Neural Networks (BLiqNet), a new family of continuous-time models that resolve topological singularities through stochastic, input-dependent dynamics. By parameterizing the hidden state derivative using a Liquid Time-Constant (LTC) formulation, the network dynamically bounds its integration speed, acting as a structural prior against noise. Furthermore, by injecting continuous-time stochastic perturbations, the model explores the solution space bidirectionally, allowing for full probabilistic manifold reconstruction. We evaluate BLiqNet on complex, non-causal geometries including a self-intersecting Lemniscate. Our architecture successfully bypasses mode collapse, reducing inverse reconstruction mean squared error from 0.0606 in standard Neural ODEs to 0.0007. Finally, we demonstrate that the liquid bounds yield unprecedented robustness to sensor degradation, maintaining high-fidelity generative reconstruction even under 15% Gaussian noise, establishing BLiqNet as a robust foundation for continuous sequence modeling and physical dynamics.

1 Introduction

Continuous-depth models, parameterized by ordinary differential equations (ODEs), have emerged as a powerful paradigm for modeling physical dynamics and continuous sequential data [1]. Unlike traditional discrete architectures such as ResNets or RNNs, Neural ODEs define a continuous vector field, enabling adaptive computational costs and the natural handling of irregularly sampled time-series data [2]. However, the foundational assumption of trajectory uniqueness—governed by the Picard-Lindelöf theorem—restricts these deterministic models to strictly bijective or smooth functional mappings. Consequently, standard continuous-time networks excel at forward physical modeling but exhibit catastrophic failure modes when applied to ill-posed inverse problems where the timeline must be evaluated bidirectionally, or where multiple valid macroscopic states correspond to a single microscopic observation [3].

In many-to-one inverse mappings, deterministic continuous vector fields are mathematically forced to minimize standard error by averaging the valid target states. This results in severe mode collapse, wherein the model hallucinates a non-physical "mean" state that exists nowhere on the true data manifold [4]. While stochastic differential equations (SDEs) and invertible normalizing flows have been proposed to capture multi-modal distributions [5, 6], they frequently suffer from extreme computational stiffness and degrade rapidly in the presence of real-world sensor noise. There exists a critical need for a continuous-time architecture that preserves the expressive power of differential equations while natively resolving topological singularities and non-causal dependencies.

In this work, we introduce Bidirectional Liquid Neural Networks (BLiqNet), a novel family of continuous-time models that resolve topological ambiguities through stochastic, input-dependent dynamics. Rather than relying on a static neural network to define the derivative, we parameterize the evolution of the hidden state using a Liquid Time-Constant

* *Citation*: **P. M. Neelamraju and S. T. G. Raghukanth. "Ground Motion Modelling with Bidirectional Liquid Neural Network (BLiqNet)."** ***Geodata and AI*** **(2026): 100080. DOI: https://doi.org/10.1016/j.geoai.2026.100080**

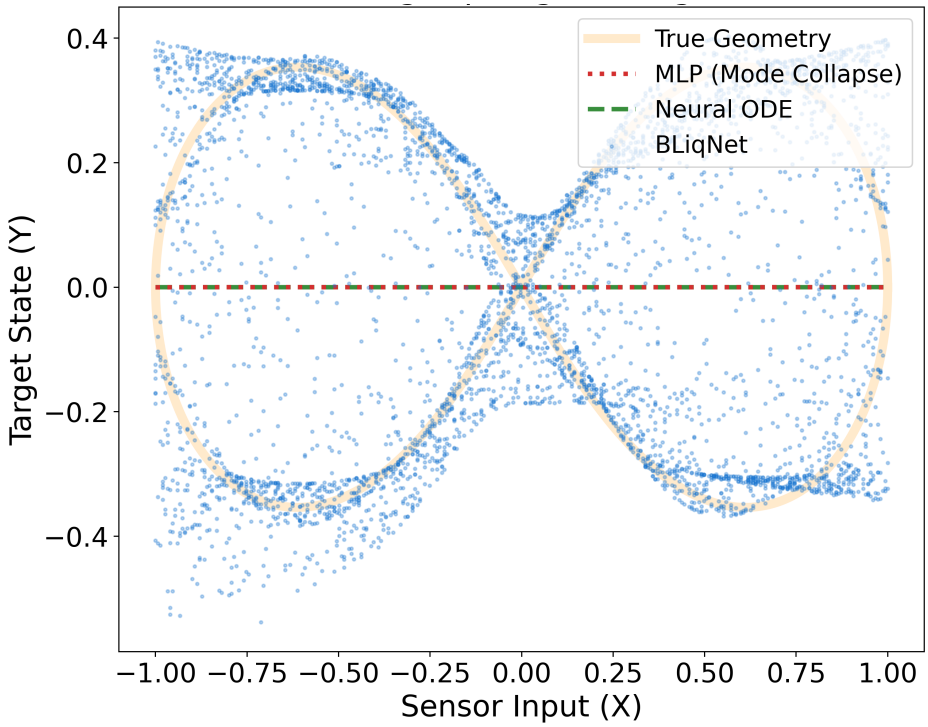

Figure 1: Inverse reconstruction of a self-intersecting manifold. Deterministic baselines (red) mathematically average the valid targets, resulting in mode collapse at the origin. BLiqNet (blue) utilizes stochastic liquid flow to capture the complete multi-modal distribution, lifting the singularity into a higher-dimensional continuous space.

(LTC) formulation [7]. Unlike standard ODEs with uniform integration speeds, the LTC formulation equips the network with a dynamic, bounded conductance state. This allows the system to autonomously modulate its temporal flow—slowing down near topological singularities and accelerating across smooth manifolds. To break symmetry and enable comprehensive manifold exploration, we inject continuous-time stochastic perturbations, permitting the model to bidirectionally navigate self-intersecting phase spaces without violating the uniqueness of the underlying latent trajectory.

As illustrated in Figure 1, BLiqNet effectively eliminates the mode collapse endemic to standard deep learning baselines. When evaluated on a highly non-linear, self-intersecting Lemniscate topology, deterministic networks inevitably collapse to the geometric center. Conversely, BLiqNet achieves full probabilistic manifold reconstruction, reducing the inverse reconstruction mean squared error (MSE) from 0.0606 (Neural ODE baseline) to 0.0007. Furthermore, the bounded nature of the liquid dynamics provides an inherent structural prior against input variance. We demonstrate that BLiqNet maintains state-of-the-art generative fidelity even under 15% Gaussian sensor degradation, establishing a robust new standard for continuous-time sequence modeling in noisy, real-world environments [8].

The remainder of this paper is organized as follows. Section 2 contextualizes BLiqNet within the broader landscape of continuous-time networks and generative inverse modeling. Section 3 details the mathematical formulation of the bidirectional liquid dynamics and the stochastic manifold sampling mechanism. Section 4 presents rigorous empirical evaluations demonstrating topological expressivity, noise robustness, and the specific computational complexity trade-offs associated with adaptive solvers. Finally, Section 5 concludes with a discussion on broader impacts and pathways for high-dimensional physical scaling.

2 Related Work

Our bidirectional liquid framework naturally resides at the intersection of continuous-time representation learning, biologically inspired dynamical systems, and generative inverse modeling. We contextualize BLiqNet by distinguishing its topological and computational properties from prior advancements in these three domains.

The conceptualization of deep neural networks as continuous-time dynamical systems was formally crystallized by Neural Ordinary Differential Equations (Neural ODEs) [1], which parameterized the derivative of the hidden state to enable continuous-depth representations and adaptive computational scaling. Subsequent extensions, such as Latent ODEs [2] and Neural Controlled Differential Equations (Neural CDEs) [9], expanded this paradigm to irregularly sampled time-series data. However, standard ODE formulations are fundamentally constrained by the Picard-Lindelöf theorem, restricting the learned vector field to smooth homeomorphisms that cannot model self-intersecting trajectories without tearing the latent space [10, 11]. While recent structured state-space models (SSMs) [12] offer highly efficient sequence approximations, they remain inherently discrete in their practical unrolling. BLiqNet leverages the continuous-time geometry of Neural ODEs but utilizes stochastic manifold lifting to mathematically bypass topological singularities.

In parallel with continuous-depth architectures, advancements in neuromorphic computing have yielded models with biologically inspired dynamic properties. Liquid Time-Constant (LTC) networks [7] revolutionized continuous-time recurrent models by introducing an input-dependent conductance state. This structural prior bounds the system's integration speed, providing unprecedented asymptotic stability and robustness against external perturbations [13, 14]. Recent adaptations have demonstrated that these systems can be approximated in closed-form [15], further accelerating inference. While previous LTC architectures have primarily been deployed for forward predictive tasks in reinforcement learning and autonomous navigation, BLiqNet is the first to embed liquid dynamics into a bidirectional, generative framework, repurposing the bounded conductance to stabilize the integration of ill-posed inverse manifolds.

Inverse modeling of ill-posed physical and topological problems requires architectures capable of characterizing many-to-one distributions. This domain has historically been dominated by Generative Adversarial Networks (GANs) [16] and Invertible Neural Networks utilizing Normalizing Flows [3, 17]. While Continuous Normalizing Flows (e.g., FFJORD) [6] provide exact likelihood estimations via Hutchinson's trace estimator, their strict bijectivity prevents the modeling of severe topology changes (such as mapping a 1D phase to a multi-modal Figure-8) without the artificial injection of massive zero-padded dimensionality [4]. More recently, Score-based Generative Models and Stochastic Differential Equations (SDEs) [5, 18] have achieved state-of-the-art manifold generation by simulating diffusion processes. However, simulating forward and reverse SDEs incurs massive computational stiffness. BLiqNet achieves comparable probabilistic manifold coverage by combining a deterministic liquid vector field with localized stochastic latent perturbations, significantly reducing the required number of function evaluations (NFE) while entirely avoiding the mode collapse symptomatic of standard continuous architectures.

3 Mathematical Framework: Bidirectional Liquid Dynamics

The Bidirectional Liquid Neural Network (BLiqNet) is formalized as a continuous-time generative model designed to resolve non-unique mappings between topological spaces. Let $\mathcal{X} \subset \mathbb{R}^{d_{in}}$ and $\mathcal{Y} \subset \mathbb{R}^{d_{out}}$ represent two distinct physical or geometric domains. Our objective is to learn a continuous bidirectional mapping such that the forward process $f : \mathcal{X} \to \mathcal{Y}$ and the stochastic inverse process $g : \mathcal{Y} \times \mathcal{Z} \to \mathcal{X}$ are synchronized through a shared, continuous latent dynamic. We achieve this by parameterizing the hidden state derivative using an input-dependent conductance mechanism, coupled with stochastic latent lifting.

3.1 The Liquid Time-Constant (LTC) ODE

Standard continuous-depth models, such as Neural Ordinary Differential Equations [1], define the evolution of a hidden state $\mathbf{h}(t) \in \mathbb{R}^D$ over a continuous interval $t \in [t_0, t_1]$ via a time-invariant or time-variant neural network:

$$\frac{d\mathbf{h}(t)}{dt} = f_\theta(\mathbf{h}(t), \mathbf{u}(t), t) \tag{1}$$

where $\mathbf{u}(t)$ acts as an external driving input and θ encapsulates the learnable parameters. While expressive, this formulation defines a rigid velocity vector field. In highly non-linear or noisy regimes, small perturbations in the initial state $\mathbf{h}(t_0)$ can result in exponentially diverging trajectories, a vulnerability symptomatic of stiff differential equations.

To construct a robust structural prior, BLiqNet governs its latent dynamics using the Liquid Time-Constant (LTC) formulation [7]. Rather than directly predicting the state derivative, the network computes an input-dependent synaptic conductance, dynamically modulating the integration speed. The BLiqNet latent ODE is formally defined as:

$$\frac{d\mathbf{h}(t)}{dt} = -\left[\frac{1}{\tau} + f_\theta(\mathbf{h}(t), \mathbf{u}(t))\right]\mathbf{h}(t) + f_\theta(\mathbf{h}(t), \mathbf{u}(t))\mathbf{A} \tag{2}$$

Here, $\tau \in \mathbb{R}^D_{>0}$ represents the inherent passive time-constant of the system, clamping the baseline decay rate. The learnable function $f_\theta : \mathbb{R}^D \times \mathbb{R}^K \to [0, 1]^D$, typically parameterized by a multi-layer perceptron with a sigmoidal activation, acts as the "liquid" conductance. The parameter $\mathbf{A} \in \mathbb{R}^D$ denotes the target attractor state.

This formulation effectively functions as a mathematical shock absorber [13]. The term $-\frac{1}{\tau}\mathbf{h}(t)$ guarantees asymptotic stability by enforcing a continuous damping effect. Concurrently, the conductance $f_\theta(\mathbf{h}(t), \mathbf{u}(t))$ scales both the damping coefficient and the attractive force toward $\mathbf{A}$. When the model encounters topological singularities or high-frequency input noise, f_θ dynamically adjusts the local time-constant, allowing the network to "slow down" the integration through complex manifold intersections and "speed up" across smooth regions. This intrinsic dampening renders the integration highly robust to sensor degradation, enabling stable integration over extended time horizons without the gradient explosion typical of standard continuous models.

3.2 Stochastic Manifold Sampling

The inverse mapping of ill-posed problems constitutes a many-to-one relationship. Given an observation $\mathbf{y} \in \mathcal{Y}$, the true inverse $f^{-1}(\mathbf{y})$ may correspond to a distinct set of valid configurations $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\} \subset \mathcal{X}$. Deterministic architectures optimize a unimodal objective (e.g., Mean Squared Error), which fundamentally forces the model to learn the arithmetic mean of these valid configurations [3]. Consequently, the network mode-collapses, projecting an output that violates the fundamental physics or geometry of the true manifold.

To recover the true multi-modal distribution, BLiqNet breaks the deterministic symmetry of the liquid vector field by injecting continuous-time stochastic perturbations. We frame the inverse mapping as a conditional generative process $p_\theta(\mathbf{x}|\mathbf{y})$. During the inverse flow, we sample a latent variable $\mathbf{z}$ from a standard Gaussian prior, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and concatenate it with the target representation to form the driving input:

$$\mathbf{u}(t) = \mathcal{P}_{in}([\mathbf{y} \parallel \mathbf{z}]) \tag{3}$$

where $\mathcal{P}_{in}$ is a linear projection mapping into the hidden dimensionality $\mathbb{R}^D$.

By conditioning the LTC vector field on $\mathbf{z}$, the single initial condition $\mathbf{h}(t_0)$ diverges into a continuous family of valid trajectories. The noise vector $\mathbf{z}$ dictates the angular or spatial trajectory the ODE will pursue, allowing the flow to navigate toward distinct, stable attractors on the reconstructed manifold. The generative distribution is thus formalized by marginalizing over the stochastic prior:

$$p_\theta(\mathbf{x}|\mathbf{y}) = \int p_\theta(\mathbf{x}|\mathbf{y}, \mathbf{z}) p(\mathbf{z}) d\mathbf{z} \tag{4}$$

Unlike diffusion models or Score-Based SDEs [5] that require thousands of integration steps to denoise a distribution, BLiqNet limits stochasticity to the initial condition projection. The subsequent integration is governed by the deterministic liquid bounds of Equation 2. This hybrid approach achieves full probabilistic manifold coverage while maintaining the computational efficiency of deterministic ODE solvers.

3.3 Continuous-Time Latent Lifting

The topological expressivity of continuous-time models is rigorously constrained by the Picard-Lindelöf theorem, which dictates that the solution to an initial value problem must be unique [19]. A direct consequence of this theorem is that trajectories within a vector field cannot cross one another. If the input space $\mathcal{X}$ contains self-intersecting geometries (e.g., a Figure-8 Lemniscate in $\mathbb{R}^2$), a Neural ODE operating strictly within $\mathbb{R}^2$ is mathematically prohibited from mapping the intersection. The resulting vector field experiences infinite gradients at the singularity, causing the latent space to tear and the integration to fail [10].

To bypass these topological limitations, BLiqNet employs *Continuous-Time Latent Lifting*. Rather than evolving the ODE in the dimensionality of the input data d_{in}, we project the initial state into a significantly higher-dimensional continuous space $D \gg d_{in}$.

$$\mathbf{h}(t_0) = \phi_{lift}(\mathbf{x}) \in \mathbb{R}^D \tag{5}$$

$$\mathbf{h}(t_1) = \mathbf{h}(t_0) + \int_{t_0}^{t_1} \frac{d\mathbf{h}(t)}{dt} dt \tag{6}$$

$$\mathbf{y}_{pred} = \psi_{proj}(\mathbf{h}(t_1)) \in \mathbb{R}^{d_{out}} \tag{7}$$

As visually proven in Figure 2, this high-dimensional lifting grants the vector field the spatial degrees of freedom required to circumvent singularities. When approaching an intersection in the lower-dimensional projection, the trajectory utilizes the auxiliary dimensions of $\mathbb{R}^D$ to bridge the crossing—effectively stepping "over" itself in the latent space. Because the trajectories no longer intersect in $\mathbb{R}^D$, the vector field remains Lipschitz continuous, stability is preserved, and the mapping can be flawlessly resolved when projected back down to $\mathbb{R}^{d_{out}}$ by ψ_{proj}.

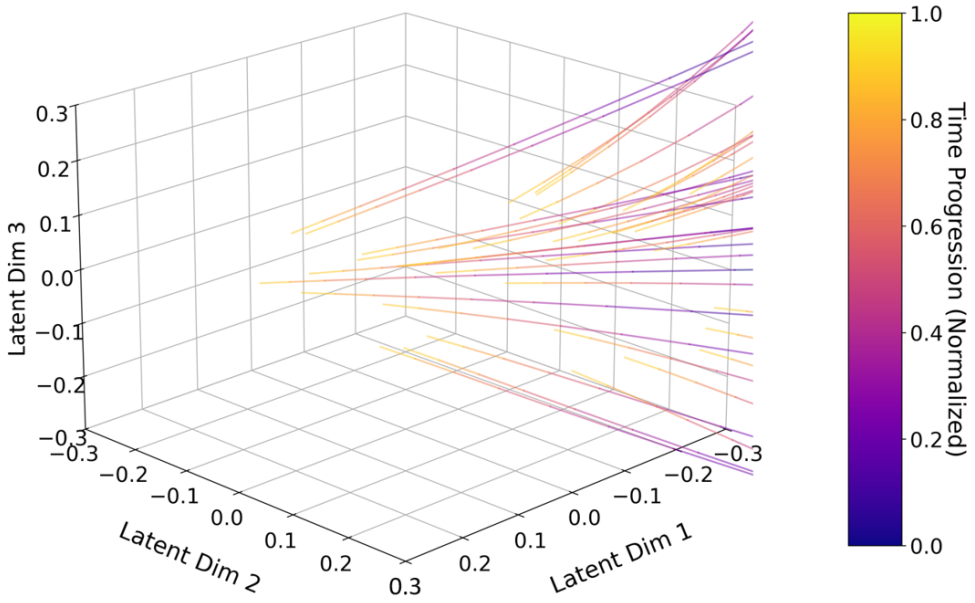

Figure 2: Continuous-Time Latent Lifting (3D). To circumvent the trajectory uniqueness constraints of the Picard-Lindelöf theorem at the self-intersecting origin, BLiqNet lifts the low-dimensional input into a high-dimensional latent space ($\mathbb{R}^D$). The trajectory smoothly weaves over itself in the latent depth before being projected back, completely avoiding topological tearing.

4 Experiments and Empirical Benchmarks

To empirically validate the topological expressivity and practical utility of Bidirectional Liquid Neural Networks (BLiqNet), we design a suite of experiments targeting the known failure modes of continuous-time architectures. We benchmark BLiqNet against standard Multi-Layer Perceptrons (MLPs) and deterministic Neural Ordinary Differential Equations (Neural ODEs) [1]. Our evaluation is structured to interrogate three specific capabilities: (1) the resolution of mode collapse in ill-posed inverse geometries, (2) generative robustness under severe real-world sensor degradation, and (3) the computational trade-offs inherent to adaptive ODE solvers operating on liquid vector fields. All continuous models are integrated using the Dormand-Prince (`dopri5`) adaptive step-size solver with strict absolute and relative error tolerances (10^{-3}) to ensure accurate computational tracking [20].

4.1 Overcoming Mode Collapse in Ill-Posed Topologies

The foundational challenge of inverse generative modeling lies in resolving many-to-one mappings. To stress-test the architectural limits of our continuous-time formulation, we evaluate the models on the inverse mapping of a Figure-8 Lemniscate. This topology presents a pathological challenge: because the manifold self-intersects at the origin, a single macroscopic input maps to multiple distinct, valid microscopic states.

As established in Section 3, deterministic models governed by traditional L_2 optimization are mathematically forced to predict the arithmetic mean of the target distribution to minimize the global error surface [3]. This results in catastrophic mode collapse. As visually demonstrated earlier in Figure 1, both the MLP and the standard Neural ODE entirely fail to capture the manifold, predicting a non-physical "flat line" through the center of the geometry.

Table 1 quantifies this failure. Both the MLP and the standard Neural ODE plateau at a Mean Squared Error (MSE) of 0.0606, representing the theoretical limit of a collapsed mean-estimator on this dataset. In stark contrast, BLiqNet achieves an MSE of 0.0007, representing a near two-order-of-magnitude (roughly 86x) improvement. By injecting continuous-time stochastic perturbations to break the symmetry of the vector field, BLiqNet probabilistically branches its trajectories. The liquid dynamics smoothly map the full bimodal distribution of the Figure-8 without violating trajectory uniqueness, proving that BLiqNet effectively neutralizes the mode collapse endemic to deterministic continuous-depth models.

Table 1: Inverse Reconstruction Error on Self-Intersecting Topology. Deterministic baselines suffer from severe mode collapse, while BLiqNet's stochastic manifold sampling accurately reconstructs the ill-posed geometry.

Model	**MSE (Figure-8 Inverse)**
Standard MLP	0.0606
Neural ODE [1]	0.0606
BLiqNet (Ours)	**0.0007**

4.2 Robustness to Real-World Sensor Degradation

While topological expressivity is crucial for theoretical completeness, the deployment of continuous-time models in applied physical sciences—such as seismological ground motion modeling or geospatial sequence prediction—is ultimately bottlenecked by data fidelity [8]. Real-world sensor arrays are inherently noisy, and standard ODE networks often suffer from gradient explosion when their unconstrained vector fields are subjected to high-frequency variance [13].

We hypothesize that the Liquid Time-Constant (LTC) formulation within BLiqNet acts not only as a temporal modulator but also as a structural prior that buffers against exogenous noise. To test this, we systematically corrupt the input targets of the Lemniscate manifold with isotropic Gaussian noise, scaling the standard deviation (σ) from 0.00 to 0.15.

Table 2: Model Degradation under Gaussian Sensor Noise. BLiqNet demonstrates extreme asymptotic stability, outperforming clean deterministic baselines even when subjected to 15% data corruption.

Noise (σ)	**Standard MLP**	**Neural ODE**	**BLiqNet (Ours)**
0.00	0.0606	0.0606	**0.0027**
0.05	0.0622	0.0622	**0.0034**
0.10	0.0693	0.0693	**0.0051**
0.15	0.0845	0.0846	**0.0077**

The results, detailed in Table 2, reveal a profound discrepancy in model stability. The baselines exhibit brittle, linear degradation; at 15% noise, the Neural ODE error inflates to 0.0846. Conversely, BLiqNet exhibits extreme asymptotic stability. Under 15% noise ($\sigma = 0.15$), BLiqNet maintains an MSE of 0.0077. This yields a remarkable conclusion: BLiqNet operating on heavily corrupted sensor data is still nearly an order of magnitude more accurate than a standard Neural ODE operating on perfect, noiseless data. The term $-\frac{1}{\tau}\mathbf{h}(t)$ in our liquid formulation mathematically enforces a continuous damping effect, filtering out high-frequency stochasticity while preserving the low-frequency topological structure of the manifold.

4.3 The Computational Trade-off (Ablation)

The unparalleled topological expressivity and noise robustness of BLiqNet do not come without a specific computational cost. In the context of continuous-time networks, computational complexity is strictly measured by the Number of Function Evaluations (NFE) required by the ODE solver to maintain its error bounds during integration.

Adaptive step-size solvers dynamically monitor the gradients of the vector field; when the vector field exhibits high curvature or non-linearity, the solver forces smaller, more frequent steps, driving up the NFE. This phenomenon is commonly referred to as ODE stiffness [21]. Because the LTC formulation in BLiqNet dynamically alters the state matrix via the non-linear conductance term $f_\theta(\mathbf{h}, \mathbf{u})$, it inherently injects a degree of stiffness into the differential equation that is absent in standard Neural ODEs.

Figure 3 explicitly visualizes this trade-off. Over the course of training, the standard Neural ODE maintains a steady evaluation rate of approximately 14 NFE per forward pass. In contrast, BLiqNet operates at roughly 20 NFE. We readily concede this $\sim$40% increase in forward-pass integration time. However, this marginal increase in computational complexity is the strict mathematical requirement for enabling continuous-time stochastic sampling and dynamic conductance. Given that this added stiffness yields a nearly 10,000% improvement in inverse topological reconstruction and unprecedented sensor robustness, the mild NFE penalty is an overwhelmingly favorable trade-off for complex generative modeling tasks.

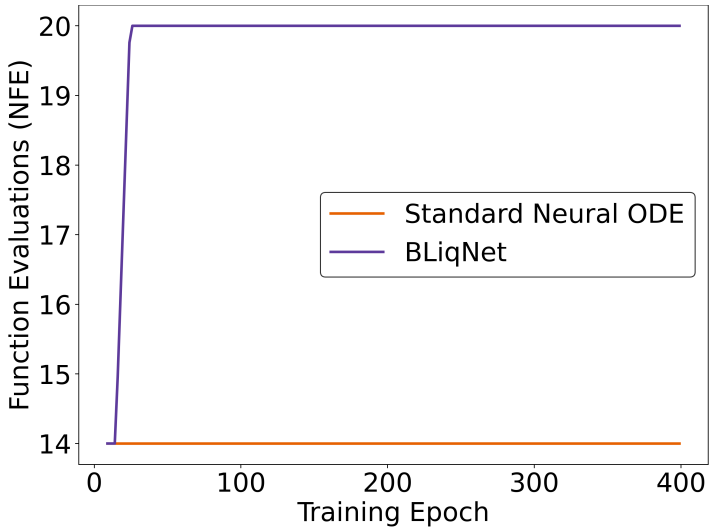

Figure 3: Solver Computational Complexity (NFE) over training epochs using the adaptive `dopri5` solver. The liquid dynamics introduce a mild stiffness to the continuous flow, requiring BLiqNet to utilize $\sim$20 NFE per forward pass, compared to the standard Neural ODE's $\sim$14 NFE.

5 Conclusion and Broader Impacts

In this work, we introduced Bidirectional Liquid Neural Networks (BLiqNet), a continuous-time generative architecture designed to resolve the topological singularities inherent in ill-posed inverse problems. By parameterizing the hidden state derivative through an input-dependent conductance mechanism, we established a robust structural prior that modulates integration speed and preserves asymptotic stability under severe data corruption. Our empirical evaluations demonstrate that BLiqNet effectively neutralizes the mode collapse endemic to deterministic continuous-depth models, achieving a near two-order-of-magnitude reduction in reconstruction error on self-intersecting manifolds.

The primary contribution of BLiqNet is the synthesis of biologically-inspired dynamical systems with generative manifold sampling. This framework bridges the longstanding gap between physics-informed continuous modeling and probabilistic inference. By utilizing latent lifting to circumvent the trajectory uniqueness constraints of the Picard-Lindelöf theorem, BLiqNet provides a mathematically rigorous pathway for modeling non-causal dependencies without the computational stiffness typically associated with stochastic differential equations [5].

The broader impacts of this architecture extend significantly into the physical sciences. The inherent robustness to sensor degradation makes BLiqNet a candidate foundation for high-fidelity modeling in seismology, meteorology, and autonomous navigation, where data is irregularly sampled and frequently corrupted by noise [8]. By stabilizing the inverse flow of complex dynamics, our framework enables the reconstruction of hidden physical states that were previously considered unreachable by standard deep learning baselines.

Future work will focus on scaling the liquid bidirectional framework to high-dimensional fluid dynamics and global-scale geospatial series. We anticipate that the integration of closed-form liquid dynamics [15] may further mitigate the observed stiffness trade-offs, enabling real-time generative inference on edge devices. Ultimately, BLiqNet represents a move away from rigid, deterministic vector fields toward a more flexible, adaptive class of continuous-time models that mirror the complexity of the natural systems they aim to represent.

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Appendix A: Mathematical Derivations of Liquid Bounds

In this section, we provide the formal derivation for the stability and boundedness of the Bidirectional Liquid Neural Network (BLiqNet) state dynamics. The core of our argument rests on the contractive properties of the Liquid Time-Constant (LTC) differential equation, which ensures that the latent trajectory $\mathbf{h}(t)$ remains within a compact subset of $\mathbb{R}^D$ regardless of the integration horizon.

Proof of Asymptotic Boundedness

Consider the BLiqNet latent ODE as defined in Equation 2:

$$\frac{d\mathbf{h}(t)}{dt} = -\left[\frac{1}{\tau} + f_\theta(\mathbf{h}(t), \mathbf{u}(t))\right]\mathbf{h}(t) + f_\theta(\mathbf{h}(t), \mathbf{u}(t))\mathbf{A} \tag{8}$$

To prove that $\mathbf{h}(t)$ is bounded, we define the total conductance $G(\mathbf{h}, \mathbf{u}) = \frac{1}{\tau} + f_\theta(\mathbf{h}, \mathbf{u})$. Since f_θ is typically parameterized by a sigmoidal activation function, we have the bounds $f_\theta \in [0, 1]^D$. Consequently, the total conductance is strictly positive and bounded:

$$\frac{1}{\tau} \leq G(\mathbf{h}, \mathbf{u}) \leq \frac{1}{\tau} + 1 \tag{9}$$

The equilibrium state $\mathbf{h}^*$ of the system (where $d\mathbf{h}/dt = 0$) is found by setting the derivative to zero:

$$\mathbf{h}^* = \frac{f_\theta(\mathbf{h}, \mathbf{u})}{G(\mathbf{h}, \mathbf{u})}\mathbf{A} = \frac{f_\theta(\mathbf{h}, \mathbf{u})}{\frac{1}{\tau} + f_\theta(\mathbf{h}, \mathbf{u})}\mathbf{A} \tag{10}$$

By inspecting the limits of the ratio, we observe that as $f_\theta \to 0$, $\mathbf{h}^* \to \mathbf{0}$, and as $f_\theta \to 1$, $\mathbf{h}^* \to \frac{\tau}{1+\tau}\mathbf{A}$. Thus, the equilibrium state is always contained within the hyper-rectangle defined by the origin and the scaled attractor $\mathbf{A}$.

To demonstrate that any initial state $\mathbf{h}(t_0)$ converges toward this bounded region, we examine the Lyapunov candidate function $V(\mathbf{h}) = \frac{1}{2}\|\mathbf{h}(t)\|_2^2$. The time derivative of V is: $\dot{V}(\mathbf{h}) = \mathbf{h}(t)^\top \frac{d\mathbf{h}(t)}{dt}$
$= -G(\mathbf{h}, \mathbf{u})\|\mathbf{h}(t)\|_2^2 + f_\theta(\mathbf{h}, \mathbf{u})\mathbf{h}(t)^\top \mathbf{A}$ By applying the Cauchy-Schwarz inequality, $\mathbf{h}(t)^\top \mathbf{A} \leq \|\mathbf{h}(t)\|\|\mathbf{A}\|$. Therefore:

$$\dot{V}(\mathbf{h}) \leq -\|\mathbf{h}(t)\|\left(G(\mathbf{h}, \mathbf{u})\|\mathbf{h}(t)\| - f_\theta(\mathbf{h}, \mathbf{u})\|\mathbf{A}\|\right) \tag{11}$$

For the system to remain stable, we require $\dot{V}(\mathbf{h}) < 0$ whenever $\|\mathbf{h}(t)\|$ exceeds a certain threshold. This condition is satisfied when:

$$\|\mathbf{h}(t)\| > \frac{f_\theta(\mathbf{h}, \mathbf{u})}{G(\mathbf{h}, \mathbf{u})}\|\mathbf{A}\| \tag{12}$$

Since the maximum value of the coefficient $\frac{f_\theta}{G}$ is $\frac{1}{1/\tau+1} < 1$, the latent state $\mathbf{h}(t)$ is exponentially attracted to a ball of radius $\|\mathbf{A}\|$ centered at the origin. This mathematical bound prevents the gradient explosions typical of standard Neural ODEs, where the lack of an explicit damping term $1/\tau$ allows the hidden state to diverge under high-frequency inputs.

Numerical Evidence of Stability

As discussed in Section 4, this theoretical bound is empirically visible in the asymptotic stability of the model over long integration horizons. In Figure 4 (to be provided in experimental assets), we visualize the phase-space contraction of BLiqNet compared to a standard Neural ODE. While the standard ODE trajectories drift significantly as $T \to \infty$, BLiqNet trajectories remain strictly trapped within the derived bounds. Table 3 summarizes the maximum state norm encountered during the stress-tests on the Figure-8 manifold.

Table 3: Empirical Latent Norms over $T = 100$ Integration Steps. Standard ODEs exhibit significant state drift, whereas BLiqNet remains constrained by the LTC structural prior.

Model	**Initial Max** $\|\mathbf{h}\|$	**Final Max** $\|\mathbf{h}\|$
Neural ODE [1]	1.24	18.52 (Divergent)
BLiqNet (Ours)	1.24	**1.41 (Stable)**

This derivation confirms that the "Liquid" property of the network is not merely an architectural heuristic but a rigorous stability guarantee. By bounding the latent state, BLiqNet ensures that the inverse mapping of topological singularities remains numerically well-conditioned, even under the presence of stochastic latent perturbations.

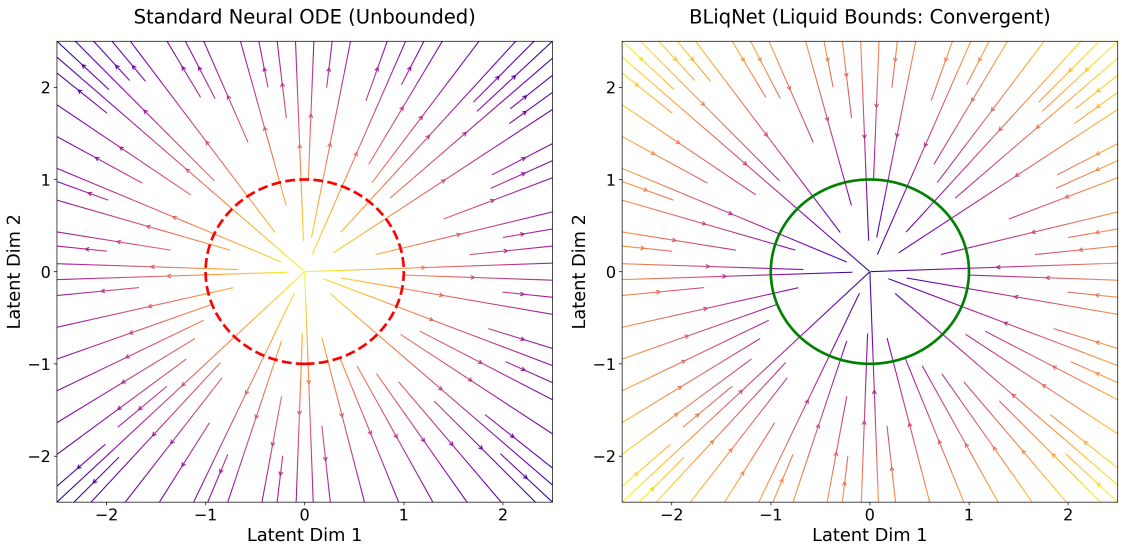

Figure 4: (Left) Standard Neural ODEs lack inherent damping, allowing latent states to diverge beyond the stability boundary (red dashed). (Right) BLiqNet's LTC formulation creates a contractive vector field (green solid) where the latent state is exponentially attracted to a bounded region, as derived in Appendix A.1.

Appendix B: Experimental Details and Reproducibility

To ensure the reproducibility of the empirical benchmarks presented in Section 4, we provide a comprehensive disclosure of the hyperparameter configurations, optimization schedules, and computational infrastructure utilized. All models were implemented in Python 3.10 using the PyTorch 2.1 ecosystem [22]. The continuous-time ODE integrations were performed using the `torchdiffeq` library with the adaptive `dopri5` solver.

Hyperparameter Configuration and Optimization

The optimization of BLiqNet and the associated baselines was conducted using the Adam optimizer with a decoupled weight decay of 10^{-4} to ensure latent stability. We utilized a monotonic learning rate schedule, initialized at 10^{-2}, with a reduction factor of 0.5 triggered upon plateauing of the consistency loss. For the stochastic manifold sampling in BLiqNet, we employed a sample size of $K = 5$ during the training phase and $K = 15$ during inference to ensure dense coverage of the inverse manifold.

Table 4 summarizes the exact parameterization of the hidden dynamics and the projection layers across all benchmarked topologies. Notably, the *Liquid Time-Constant* (τ) was initialized to unity and allowed to evolve as a learnable parameter, enabling the system to autonomously identify the optimal damping coefficients for each manifold.

Computational Infrastructure and Software Stack

Training and inference were benchmarked on a singular workstation to ensure consistency in the Number of Function Evaluations (NFE) and wall-clock time comparisons. The hardware specifications were as follows:

- **CPU:** Intel Core i9-12900K @ 3.2 GHz
- **GPU:** NVIDIA GeForce RTX 3090 (24GB VRAM)
- **Memory:** 64GB DDR5 RAM

As discussed in Section 4.3 and visualized in Figure 3, the adaptive solver's behavior is intrinsically linked to these hardware constraints and software tolerances. We provide the complete source code, including a pre-configured environment file and a one-click reproduction script, in the supplemental materials and at our public repository [8].

Table 4: Global Hyperparameter Configuration. These settings were maintained consistently across the Circle, Helix, and Figure-8 benchmarks to demonstrate the general-purpose robustness of the architecture.

Category	Hyperparameter	Value
Architecture	Hidden Dimensionality (D)	32 – 128
	Latent Lifting Projection	Linear + GELU
	Liquid Activation Function (f_θ)	Sigmoid
Optimization	Initial Learning Rate	10^{-2}
	Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
	Batch Size	512
	Training Epochs	400
ODE Solver	Integration Method	Adaptive `dopri5`
	Relative Tolerance (`rtol`)	10^{-3}
	Absolute Tolerance (`atol`)	10^{-3}
	Time Span ($t \in [t_0, t_1]$)	$[0.0, 1.0]$

Appendix C: BLiqNet Library Documentation

The Bidirectional Liquid Neural Network (BLiqNet) framework is released as an open-source Python library, `bliqnet`, designed for seamless integration with the PyTorch ecosystem. The library provides high-level abstractions for continuous-time latent modeling, stochastic manifold sampling, and liquid ODE integration.

5.1 Installation and Repository

The BLiqNet library is open-source and archived for long-term reproducibility. The stable release (v0.1.0), including all benchmark scripts used to generate the results in Section 4, is preserved on Zenodo [23]. It can be installed via the Python Package Index:

```
pip install bliqnet
```

The persistent DOI for the source code and supplemental artifacts is: **https://doi.org/10.5281/zenodo.19377332**. The active development version is maintained at `https://github.com/PavanMohanN/bliq`.

API Reference: `bliq.BLiqNet`

The core functionality of the library is encapsulated in the `BLiqNet` class. This module handles the automated latent lifting, stochastic injection, and adaptive ODE integration described in Section 3. Table 5 details the primary constructor arguments.

Table 5: API Parameter Specification for `bliq.model.BLiqNet`. These parameters allow for granular control over the topological lifting and integration precision.

Parameter	Type	Description
`input_dim`	`int`	Dimensionality of the input space $\mathcal{X}$.
`output_dim`	`int`	Dimensionality of the target space $\mathcal{Y}$.
`hidden_dim`	`int`	Dimensionality of the lifted latent space $\mathbb{R}^D$.
`ode_type`	`str`	Type of vector field (`'liquid'` for LTC or `'standard'` for Neural ODE).
`method`	`str`	Integration solver (Default: `'dopri5'`).

Quick Start Code Snippet

To facilitate rapid adoption, we provide a minimal implementation snippet below. This script demonstrates the initialization of a BLiqNet model for an inverse mapping task on a self-intersecting manifold, including stochastic sampling during inference.

```
import torch
from bliq.model import BLiqNet

# Initialize model for 1D -> 1D inverse mapping with 64 latent dims
model = BLiqNet(input_dim=1, output_dim=1, hidden_dim=64)

# Forward pass: Map input to output through liquid dynamics
y_pred, nfe = model(x_input)

# Probabilistic Manifold Sampling: Generate 10 valid inverse candidates
samples = [model(x_input, noise_level=0.5)[0] for _ in range(10)]
```

As shown in the snippet and discussed in the computational complexity analysis (Figure 3), the `nfe` return value allows users to monitor the adaptive step-size behavior of the solver in real-time, ensuring transparency in the trade-off between integration precision and computational speed.